

Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-3138
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. This document records the testing approach, tools used, and results obtained for the ComicCraft application.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	FastAPI endpoints: / , /generate (form), /generate-comic/json, /test-image, /export-success
Test Environment	Localhost / Local Development Server (uvicorn app.main:app, http://127.0.0.1:8000)
Test Date	01 October 2026

Step 2: Test Scenarios

Comic generation is heavy: each request calls Gemini Flash, Gemini Pro, Stable Diffusion (5 panel images) and the FPDF exporter. Virtual-user counts are therefore kept low for generation scenarios and higher for lightweight pages.

S.No	Test Scenario	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate one comic from a story prompt (e.g. brave fox in an enchanted forest, dramatic, anime) via POST /generate	1	300	Comic preview page returned with 5 panels and PDF path, without errors
2	Generate comics with different tone and art style combinations (funny / comic book, poetic / pixel art) via POST /generate and /generate-comic/json	2	300	API handles multiple requests successfully; each returns valid layout data
3	Multiple users generate comics simultaneously	3	600	System remains responsive; no crashes, timeouts or out-of-memory errors

S.No	Test Scenario	No. of Virtual Users	Duration (sec)	Expected Outcome
4	Load on lightweight routes: home page (/), /docs, /export-success	20	60	Pages load quickly while no generation is running; very low error rate

Step 3: Performance Test Results

Run the provided locustfile.py, then copy the values from the Locust Statistics tab and Task Manager / nvidia-smi into the Actual Value column and set Pass or Fail.

S. No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg) - GET /	< 1 second	[Fill after running Locust]		Home page load time
2	Response Time (Avg) - POST /generate	< 120 seconds	[Fill after running Locust]		Full pipeline: outline, story, 5 images, layout, PDF
3	Response Time (Max) - POST /generate	< 300 seconds	[Fill after running Locust]		Slowest comic generation observed
4	Throughput (Req/sec)	> 0.5 req/sec (overall)	[Fill after running Locust]		Mostly lightweight routes; generation is much slower per request
5	Error Rate	< 5%	[Fill after running Locust]		Check Failures tab (Gemini rate limits, timeouts)
6	CPU Utilization	< 80%	[Fill after running Locust]		Stable Diffusion is the main consumer
7	Memory Utilization (RAM / GPU VRAM)	< 80%	[Fill after running Locust]		Model stays loaded in memory

Step 4: Observations & Analysis

Key Findings:

Summarise your real results here, for example: average and maximum response time of /generate, error rate, and CPU / memory usage during the test. Mention whether the application stayed responsive when 2 to 3 users generated comics at the same time.

Bottlenecks Identified (expected, confirm with your results):

- Stable Diffusion image generation is the slowest step, because 5 panel images are created one after another; on a CPU-only machine this takes much longer than on a GPU.
- Gemini Flash and Gemini Pro calls depend on network latency and API rate limits, which can cause slow responses or occasional errors under concurrent requests.
- Concurrent generation requests compete for the same CPU/GPU and memory, so response time grows with the number of simultaneous users.
- FPDF export and layout building take very little time compared with the AI steps.

Optimization Steps Taken:

- Used FastAPI with Uvicorn for efficient API handling.
- Loaded the Stable Diffusion pipeline once and reused it for every panel instead of reloading per request.
- Kept the outline, story, image, layout and export steps modular so each can be tuned independently.
- Added try-except blocks and HTTPException handling to fail gracefully on API or JSON errors.
- Tested with an increasing number of virtual users (1, 2, 3, then 20 on light routes) to verify stability.
- Further options: use a GPU with fp16, reduce inference steps or image size, run image generation in a background job queue.

Step 5: Screenshots / Evidence

Panel 1: The Discovery



Laki wanders through a shadowy alleyway in a towering futuristic city. She notices a glowing, metallic pen resting on a discarded crate, pulsing with an unnatural azure light.

Caption: THE DISCOVERY

In the shadow of monolithic skyscrapers, where the city's neon pulse fails to reach, secrets are often left to rust.

Dialogue:

Laki: What is that... it's calling to me.

Panel 2: First Stroke



Curiosity overriding her fear, Laki kneels on the metallic ground and uses the pen to draw a simple bird in the air, watching in awe as the ink solidifies into glowing light.

Caption: FIRST STROKE

Reality bends to the touch of the impossible. Every stroke bleeds raw energy into the cold air.

Dialogue:

Laki: It feels so warm... almost alive.

AI COMIC CREATOR

ComicCraft

Turn your imagination into a complete AI-powered comic book.

01 Your Story

Story Prompt

A young girl discovers a magical pen that brings her drawings to reality.

Main Character

Laki

Setting

 Futuristic City

02 Creative Style